

HN-01 · SPORTS NUTRITION · HUMAN NUTRITION

HS-AminoForce 4:1:1

L-Leucine, L-Isoleucine, L-Valine (4:1:1)

INCI	CAS	EC	FAMILY	SUPPLIED AS
L-Leucine, L-Isoleucine, L-Valine (4:1:1)	61-90-5 +2	—	HN-01	Instantized powder

01 POSITIONING

The recovery claim of the sports aisle — in the ratio and the solubility the brand specifies.

02 HOW IT WORKS

- 01 Leucine trigger
Leucine activates the mTOR pathway, the switch for muscle-protein synthesis.
- 02 Direct muscle fuel
BCAAs are metabolized in muscle rather than liver — usable energy during training.
- 03 Ratio engineering
4:1:1 maximizes the leucine trigger; 2:1:1 balances the full recovery profile.

03 APPLICATIONS

- Intra-workout beverages
- Recovery powders
- Ready-to-drink sports drinks
- Amino stick packs
- Vegan protein fortification

04 FORMULATION GUIDE

USE LEVEL	5–10 g serving
PH & CONDITIONS	Stable across beverage pH; instantized grade wets without foaming
— Instantized (lecithin-treated) grade avoids the floating-clump problem in cold water.	
— Mask bitterness with acid/sweetener systems; pair with electrolytes for intra-workout.	

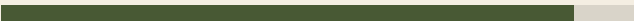
05 TYPICAL SPECIFICATION

Appearance	White instantized powder
Ratio	2:1:1 / 4:1:1 (L:I:V)
Assay (total BCAA)	≥ 99%
Source	fermentation, plant-based

Typical parameters. The specification on the current TDS and the per-lot Certificate of Analysis govern.

06 PERFORMANCE — DIRECTIONAL

Muscle-protein synthesis signal (class data)
BCAA 4:1:1



Instantized solubility
Instantized grade



Values are representative of the ingredient class, based on published technical literature, and shown directionally for comparison. Final performance must be validated in the customer's formulation.